Fiscal Year:	FY 2019	**Task Last Updated:**	FY 04/14/2020
PI Name:	Stamps, Blake Ph.D.		
Project Title:	Biodeterioration and Biocorrosion in Spaceflight Ecosystems: Implications for Material/ Microbiome Interactions on the International Space Station		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80401	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016 SLOANMOBE2016. Sloan ISS MoBE Postdoctoral Fellowship - Post-Doctoral Fellowships in Space Biology to Study the Microbiome of the ISS as a Built Environment: Using ISS as a Microbiological Observatory
Start Date:	10/01/2017	**End Date:**	09/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
Contact Monitor:	Wolf, Yolanda	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Spear, John Ph.D. (MENTOR/ Department of Civil and Environmental Engineering, Colorado School of Mines)		
Grant/Contract No.:	Sloan G-2017-9853		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP NOTE: This project is funded by the Alfred P. Sloan Foundation, as a result of a Space Act Agreement between NASA and the Sloan Foundation, for Collaborative Research Proposal for Microbiology of the Built Environment (MoBE). The purpose of this Space Act Agreement is to enhance the ability of NASA and the Sloan Foundation to support scientific research and the dissemination of information about the effect of spaceflight on microbial ecosystems in spacecraft such as the International Space Station (ISS) through sponsorship of postdoctoral fellowships. NASA is providing the isolates for all studies. Corrosion represents a significant cost to the global economy, representing up to 6% of US GDP. An unknown percentage of this corrosion is due to the interaction of microorganisms and materials resulting in microbially influenced corrosion (MIC). Microorganisms capable of MIC are phylogenetically diverse, and found in a broad diversity of environments. The International Space Station (ISS) is no different than many Earth-bound built environments and the potential exists that corrosive organisms are present on the ISS or any other spaceflight system. Broadly this proposal seeks to characterize the expansive set of microbial isolates obtained from the ISS over the past 15 years using a combination of physical, microscopic, and electrochemical means to understand the capability and rates of MIC from space-adapted microorganisms. Concurrently isolates will also be assessed for their ability to degrade polymeric coatings, specifically polyurethane (PU). Coatings are at risk for microbial degradation and may then expose metallic surfaces for additional corrosion. Again, microorganisms from the ISS culture collection will be used to determine the risk of biodegradation using a PU analog, Impranil. Finally, to understand if isolates have obtained any differential ability to corrode or degrade coatings near relatives will be purchased from culture collections and assessed for their ability to corrode metallic surfaces and degrade PU. The results will be used to direct detection and mitigation efforts on future spaceflight systems and ensure the reliability and safety of human inhabited spaceflight vehicles in the years to come.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results will be used to direct detection and mitigation efforts on future spaceflight systems and ensure the reliability and safety of human inhabited spaceflight vehicles in the years to come.
Task Progress:	The funding graciously provided by the Sloan Foundation was split between two major microbial activities that may impact materials on the International Space Station (ISS)-biodegradation and biocorrosion. The proposed work began October 1, 2017 and concluded fall 2019. I now have a research scientist position with UES, Inc., the primary contractor for the Air Force Research Laboratory (AFRL) in Dayton, Ohio. Our plan was ambitious, and despite the slightly truncated timeline major activities were completed through the funded period. This report summarizes major accomplishments and work remaining to complete. A total of 120 microbial cultures were obtained from NASA that were successfully propagated and maintained over the fellowship. Of the 120 tested isolates 35 displayed some amount or capability of polyurethane degradation with three showing the greatest amounts of polymer degradation potential. To further link the observed phenotype to a genotype that could be useful to NASA, 14 isolates were selected for DNA sequencing. DNA was extracted from a pellet of cell mass preserved from the initial culture received from NASA, and both Illumina MiSeq and Oxford Nanopore MinION sequencing were performed. Final annotation is underway and we expect to release the data publicly alongside the manuscript in preparation in late Spring 2020. In January 2019 a large-scale corrosion test was begun to compliment the biodegradation work. Originally two time points were planned, and two materials (stainless steel and Al2024 aluminum). Final results are pending, but initial corrosion calculations suggest differences of over an order of magnitude exist between the highest corroders and sterile controls, although all conditions are considered low corrosion risk (<1 mil per year, or MPY), though these risk estimates are for oil and gas pipelines, not spacecraft. Work continues in collaboration with the Air Force Research Laboratory to bring the story to a close. Through new and existing collaborations at AFRL, I was able to secure imaging and profilometry of approximately 100 aluminum coupons, including both culture exposed and controls. This represents over 500 hours of instrument and technician time that I will leverage at no cost to bring an extremely high-resolution understanding of the corrosion dynamics of ISS built-environment isolates. Three coupons from each culture and control condition were chosen based on maximum corrosion rates to provide the most extreme examples of potential surface pitting. While we were unable to complete additional experimentation using stainless steel, or investigate possible differences in corrosion via electrochemical corrosion cells as originally planned, the work still stands to contribute significantly to our understanding of common skin and built- environment associated microorganisms and their impact on materials corrosion and biodegradation of the ISS built environment. The fellowship has also allowed me to complete multiple manuscripts, with more in preparation, including the built environment of the Orange County Water District as well as multiple studies focused on geomicrobiology including Mono Lake, California. Another was the result of interactions with a fellow postdoctoral fellow at the Colorado School of Mines, and another the culmination of an intense built-environment collaboration with the Air Force Research Laboratory. I was also fortunate enough to mentor several graduate students during my time at Colorado School of Mines – I am a co-author on one student's manuscript. I was able to complete five (two first author, three co-authored) manuscripts as a result of this fellowship, and after final publication, the fellowship will have directly resulted in two first-authored, two other first-authored, and numerous co-authored manuscripts that would not have been possible without the support of the Sloan Foundation. [Ed. note 4/13/2020: compiled from PI's final report submitted to the Sloan Foundation]
Bibliography Type:	Description: (Last Updated: 10/09/2024)

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